

Aarya Venkat

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EDUCATION

UNIVERSITY OF GEORGIA

PHD BIOCHEMISTRY
ARCS Foundation Scholar
Future Faculty Fellow
Dec 2023

UC, SAN DIEGO

MS CHEMISTRY
June 2017

SKILLS

PROGRAMMING

Python • CSS • Git • Bash • Docker •
Kubernetes • Argo Workflows • Nextflow
• Cloud Computing (AWS, GCP)

BIOINFORMATICS

Sequence Analysis

Metagenomic sourcing • Sequence
alignment • Intron/Exon analysis •
Clustering • ESM inference

Molecular Dynamics

MDAnalysis • Amber suite • GROMACS •
Schrodinger • Forcefield parameterization

Molecular Modeling

Structure-prediction • Rosetta •
Molecular Docking • VMD • PyMOL •
Homology modeling • RFDiffusion

Deep Learning

Explainable AI • Protein language models
• Dataset preparation • App/Tool
development • Predictive modeling

RESEARCH

Writing • Editing • Analytical skills • Data
Visualization • Figure generation •
Illustrator • Multivariate Statistical
Analysis • Making presentations

MISCELLANEOUS

Leadership • Public speaking •
Documentation • Systems administration

AWARDS

2023 | Eriksson Lecture Scholarship
2023 | ASBMB Conference Travel Award
2022 | Glycobiology Poster Award
2022 | Glycobiology Travel Award
2021 | Glycobiology Poster Award
2021 | Glycobiology Travel Award
2021 | UGA Travel Award
2021 | ARCS Foundation Scholar
2021 | Three Minute Thesis Award

RECENT EMPLOYMENT HISTORY

GINKGO BIOWORKS, INC | PROTEIN ENGINEER III

Dec 2023 – current | Boston, MA

- Performed protein engineering campaigns by combining molecular docking, modeling, inference with evolutionary models like ESM2 and EVCouplings, sequence-based approaches like metagenomic sourcing coupled with structure-prediction tools like ESMFold, AlphaFold2, and Boltz-1 to design and enhance protein function objectives like specificity, thermostability, activity, and fidelity in high-budget contracts.
- Subject matter expert and tech liaison for multiple projects involving AI, protein variant design, or general computational biology expertise. Engaged and led customer-facing meetings, explaining the research performed, benefits, risks, and expectations for future research.
- Developed PyTorch deep learning models fine-tuned from ESM2, using Low Rank Adaptation (LoRA) to build custom enzyme family-specific models. These models were used to select enzymes for Ginkgo's enzyme custom plates service, which allows a biopharma company to screen against enzymes that may catalyze a specific reaction on substrates of interest.
- Built Docker pipelines on Kubernetes for tools like RFDiffusion-AA, proteinMPNN, and proprietary tools used for large-scale protein design.

UNIVERSITY OF GEORGIA | GRADUATE RESEARCH ASSISTANT

July 2018 – Dec 2023 | Athens, GA

- Performed bioinformatics and evolutionary analyses, including molecular dynamics, phylogeny, modeling, quantum chemistry, and sequence analyses.
- Developed deep learning models and tools to classify enzyme function.
- Maintain data storage, equipment, purchase orders, protocol documentation.
- Authored over 17 manuscripts for publication, with key involvement in writing, editing, data generation, analysis, visualization, and funding.

SELECTED PUBLICATIONS

Papers (Total: 21) (citations: 354) (H-index: 10)

Venkat, A. (2025). GapClean, a tool for cleaning and improving comprehension of protein sequence alignments. In Prep.

Venkat, A., et al. (2025). Glydentify, a deep learning tool for classification of glycosyltransferase function. In Prep.

Venkat, A. et al. (2023). Mechanistic and evolutionary insights into isoform-specific 'supercharging' in DCLK family kinases. eLife.

Aceil, J.†, Venkat, A.†, et al. (2023). Prevalence and homology of the pneumococcal serine-rich repeat protein at the global scale. Microbiology Spectrum.

Meng, Y. et al. (2023). Phosphorylation-dependent pseudokinase domain dimerization drives full-length MLKL oligomerization. Nature Communications

Venkat, A., et al., (2022). Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. Journal of Biological Chemistry.

Taujale, R., Venkat, A., et al. (2020) Deep evolutionary analysis reveals the design principles of fold A glycosyltransferases. eLife.

INVITED TALKS

2023

Evolution of fold-A glycosyltransferases and GTXplorer, a new tool for comparative glycomics. ASBMB Conference. Seattle Conference Center

Evolution of functional diversity of fold A glycosyltransferases. Eriksson Lecture. Complex Carbohydrate Research Center

2022

AlphaFold2: protein structure-prediction in the modern era. BCMB3600. University of Georgia

Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. Southeast Enzyme Conference

2017

Principles of chemical equilibria, UC, San Diego

WORKSHOPS AND SYMPOSIA

2023

AI and Deep Learning, Applications in Bioinformatics

2022

Protein Engineering Symposium - Chair

2021

Unfolded Protein Response Symposium - Chair

2020

Writing a Diversity Statement - GradTeach Workshop - Workshop leader, 2020

Spring Teaching Symposium - Workshop leader

Spring Teaching Symposium - Organizer

2019

Cancer Immunotherapies Symposium - Organizer

SERVICE

NATIONAL HISTORY DAY JUDGE

2020-2022 | Athens, GA

Judged Performances and Documentaries for middle school and high school students. Provided critiques to aid them in state and national competitions.

SCIENCE OLYMPIAD LAB MANAGER

2017-2020 | Athens, GA

Aid in setting up Chemistry and Forensic labs for high school Science Olympiad. Ensure lab safety protocols are followed as students compete.

PRESENTATIONS/POSTERS

2023

Glydentify, a deep learning tool for classification of glycosyltransferase function. Society for Glycobiology. Big Island, HI

2022

Modular evolution of fold-A glycosyltransferases and new tools to analyze them. Society for Glycobiology. Fernandina Beach, FL

Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. Glycoscience Training Program. Athens, GA

2021

Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. Society for Glycobiology.

2020

Playground Learning: Team Learning and Gamification. USG Teaching & Learning Conference (Cancelled: COVID-19)

Unfolded Protein Response. Biochemistry Symposium Chair. University of Georgia.

2019

Deep Evolutionary Analysis Reveals the Design Principles of Fold A Glycosyltransferases. Glycobiology Conference Poster. PI: Natarajan Kannan, UGA

Cancer Immunotherapies. Biochemistry Symposium Host. University of Georgia.

Teaching and Laboratory Assistant Orientation. Delivered Lecture on Efficient Grading Practices. University of Georgia.

2018

Teaching and Laboratory Assistant Orientation. "Teaching Tips" Q&A Panelist. University of Georgia.

2017

PathInsight: A Novel Tool for Modeling Biomolecular Pathways. Thesis Defense. UCSD.

Guest Lecture on Chemical Equilibrium. Chemistry 6B, UCSD.

2016

Does Competition Enhance Learning Over a Relaxed Guided Lesson? Teaching Methods Poster Presentation. Natural Sciences Building, UCSD.